Jaemin Park

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EDUCATION

Seoul National University, *Dept. of Computer Science and Engineering* | Seoul, Korea — 2022 - Current
GPA: 3.41 / 4.3 Major GPA: 3.46 / 4.3

Gyeonggi Science High School | Suwon, Korea GPA: 4.03 / 4.3 — 2019 - 2021

Major Courses: Computer Programming (OOP) | Principle of Programming (FP) | Automata Theory | Data Structure | Algorithm | Computer Architecture | System Programming | Introduction to Modern Cryptography | Modern Algebra | Introduction to Mathematical Analysis

ACHIEVEMENTS

Competitive Programming

The 48th Annual World Finals of the International Collegiate Programming Contest - 9th place (Bronze Medal) - Team *NewTrend* — 2024
The 2024 ICPC Asia Pacific Championship - 1st place (Champion) - Team *NewTrend* — 2024
The 2023 ICPC Asia Hue City Regional Contest - 1st place (Champion) - Team *NewTrend* — 2023
The 2024 ICPC Asia Seoul Regional Contest - 10th place - Team *aespa* — 2024
The 2023 ICPC Asia Seoul Regional Contest - 3rd place - Team *NewTrend* — 2023
The 2022 ICPC Asia Seoul Regional Contest - 4th place - Team *NewTrend* — 2022
Samsung Collegiate Programming Cup 2023 - 2nd Prize — 2023
Samsung Collegiate Programming Cup 2022 - 4th Prize — 2022
Korea Selection Test for International Olympiad of Informatics 2021 - 6th place — 2021
Korean Olympiad of Informatics 2021 - Silver Prize (6th place) — 2021
Korean Olympiad of Informatics 2020 - Silver Prize (5th place) — 2020
Korean Olympiad of Informatics 2019 - Silver Prize — 2019
Nexon Youth Programming Challenge 2021 - Bronze Prize (4th place) — 2021
Nexon Youth Programming Challenge 2020 - Bronze Prize (8th place) — 2020
Reply Code Challenge Teen Edition - 4th place — 2021
Codeforces - Rating 2715 (International Grandmaster)

Research & Hackathon Awards

Nexon Young Programmers Cup 2026 — Master Track - Bronze Prize (9th place) — 2026
2026 Korea Quantum Information Competition - Director's Award (3rd place) — 2026
Samsung Humantech Paper Award 2021 Mathematics / Data Processing section - Gold Prize — 2021
Samsung Humantech Paper Award 2020 Mathematics / Data Processing section - Bronze Prize — 2020

EXPERIENCE

Huawei Technologies Co., Ltd. — Hong Kong Research Center — 2025
Research Intern

- Worked on an LLVM-based C++ framework for optimizing large-scale tensor computation graphs.
- Designed and implemented a new compiler optimization pass, significantly improving its runtime through efficient algorithms and data structures; merged into the project's codebase.
- Developed an efficient DAG analysis tool leveraging flow-based graph algorithms to compute structural metrics and evaluate optimization passes.

Korean IOI Training Camp, Teaching Assistant — 2021 - 2024

- Assisted in teaching various topics on algorithms to 50+ candidates of Korean IOI delegate

PROJECTS

Reinventing the Wheel — Source code

- Developed a hardware simulation platform that makes the internal behavior of CPUs, memory systems, and I/O devices observable, from individual logic gates to complete systems executing programs.
- Implemented ALUs, memory models spanning flip-flops, SRAM, and DRAM, cache hierarchies, and RV32I/RV32IM processors, integrating memory-mapped I/O to run compiled C game workloads. Future directions include AI accelerators and general-purpose GPUs.
- Designed the platform for computer architecture research, enabling controlled comparisons of processor designs, cache policies, and memory technologies under realistic gate and device timing models.
- Built an event-driven C++20 engine with structural and behavioral models, gate-derived delay tables, and graph-based computation of safe clock periods.

RESEARCH

Data Structure for Maintaining Kth Maximum / Minimum Element of Polynomial Functions 2021

- Developed an efficient algorithm that can maintain Kth maximum / minimum element of linear functions, with geometric transformations using Modified Quad Tree (MQT) and Persistent Angular Sorting (PAS) data structure.
- Generalized PAS to arbitrary polynomial functions, and implemented MQT, PAS with buckets to show its efficiency.

Competitive Reinforcement Learning for Graph Coloring Games 2020

- Devised a 2-player graph game in order to identify vertices important in graph connectivity, and found optimal strategies using reinforcement learning.
- Applied multi-agent reinforcement learning with DQN, DDQN, D3QN, and tested its efficiency in big graphs.

New Algorithm for Maintaining Lower/Upper Envelope of Similar Convex Figures 2020

- Improved the Li-Chao Tree data structure to maintain the lower / upper envelope of quadratic functions.
- Proved that the improved algorithm can be applied to similar convex figures, and implemented it with buckets and divide & conquer methods to show its efficiency.

New Techniques for Dynamic Programming Optimization Using Convex Hull and Segment Tree 2019

- Implemented a dynamic convex hull data structure using splay trees, in order to optimize particular recurrences that can be reduced to finding extreme points in convex hull.
- Devised Quad Li-Chao Tree data structure and a bottom-up segment tree with convex hulls and tested its efficiency.

SKILLS

Languages C/C++ (Fluent), Java (Intermediate), Python (Intermediate), Scala (Basic)